

RadicalTrainingPlatform: Platform Plan (spec-first)

Basis: audit of `main` @ 2b17f51. Companion file: the Architecture Audit. Assumptions marked **[A]** are awaiting Roger’s answers in audit section 4.

Part 1. Product spec

One-line: a cross-platform certification study platform. Practice questions, blueprint tracking, spaced repetition and an interactive lab simulator, vendor-neutral, starting with Nutanix.

Users: an individual studying for a certification, on a laptop or phone, often offline.

Principles 1. Web first **[A]**: the PWA is the product, Desktop is an optional shell. 2. Content is data: banks compile to versioned JSON, with a schema, sources and an exam version on every question. 3. Local-first: all progress stored on the device, export and import included. No account required. 4. One simulator, one source, consumed by every shell. 5. Every claim in the README is backed by a CI check.

Part 2. Requirements (testable)

ID	Requirement	Acceptance
RTP-C-01	Questions compile from markdown to <code>exams.json</code> with schema validation	CI fails on schema error
RTP-C-02	Each question carries <code>examVersion</code> , <code>blueprintObjective</code> , <code>source</code>	Linter rejects missing fields
RTP-C-03	<code>errata.json</code> is applied at build time and lists every known key correction	Test asserts Q35 = D; all 3 documented corrections present
RTP-C-04	Question and exam counts in README/CLAUDE.md are generated	CI diff check
RTP-S-01	Study, practice and timed-exam modes	E2E tests per mode
RTP-S-02	Progress per question stored locally (IndexedDB on web, SQLite on desktop)	Survives restart; export/import round trip
RTP-S-03	Spaced repetition (SM-2) scheduling	Unit tests on scheduler
RTP-S-04	Blueprint coverage and weak-area view	Matches per-objective stats
RTP-S-05	Multi-select, ordering and lab-scenario question types	Parser and UI tests
RTP-L-01	Single simulator package, consumed by Web and Desktop	No duplicated JS in repo; CI check
RTP-L-02	Simulator scenarios have pass/fail checks and are linked to blueprint objectives	Scenario tests
RTP-P-01	PWA installable and fully offline after first load	Lighthouse PWA check in CI
RTP-P-02	Accessibility WCAG AA on study views	axe check in CI
RTP-Q-01	Core test coverage at least 70%, gated	CI threshold
RTP-Q-02	Desktop smoke test (launch, load an exam)	Headless run in CI
RTP-R-01	Tagged releases with changelog and signed artifacts	Release workflow
RTP-G-01	LICENSE present and content/trademark posture documented	File exists; legal note in README

Nutanix scope today and beyond

Nutanix is the first vendor and the only one built: NCA-75, NCP-US 6.10, NCP-CI 6.10, NCP-AI 6.10 and NCM-MCI 6.10 (1,598 questions), plus cheat sheets, a cross-exam comparison, an NCM-MCI lab walkthrough, and a 49-view Prism Element/Central/Objects/CI/AI simulator. Roger said “started nutanix but we can do so much more” (confirmed by Roger as the simulator and the exam cheat sheets in this repo). This plan treats Nutanix as the pilot vendor and makes the platform vendor-neutral by design (Phase 5).

Part 3. Target architecture

<code>content/</code>	markdown banks (authoring) + <code>errata.json</code> + schema
<code>-> tools/build-content</code>	<code>-> dist/exams.json</code> (versioned, validated)
<code>packages/core-web</code>	TypeScript: parser consumers, session engine, SM-2, storage
<code>packages/simulator</code>	the one lab simulator (moved from Web/ and Legacy/)
<code>apps/web</code>	PWA (study + simulator), deployed to Pages
<code>apps/desktop</code>	Avalonia shell, hosts the same web build in a WebView
<code>src/Core (.NET)</code>	retained for desktop-native features and PDF export
<code>legacy/winforms</code>	frozen, excluded from CI gates

Decision to confirm: keep .NET Core for desktop and PDF export while the web gets a TypeScript session engine, or move session logic to one place. Recommended: share one engine by hosting the web build inside Desktop, and keep .NET Core only for PDF export and the legacy shell.

Part 4. Roadmap

Phase 0: Repair (1 week) - Fix mixed Avalonia package versions; add a Desktop smoke launch to CI. - Add LICENSE; decide on vendor PDFs. - Triage Dependabot: merge safe bumps, close the rest. - Regenerate README, CLAUDE.md and sprint plan from reality (1,598 questions, .NET 10, Avalonia). Mark real sprint status. - Wire `errata.json` into the parser; add the missing third correction; test it. - Do not merge `chore/external-audit`; delete it after review.

Phase 1: One simulator (1 to 2 weeks) - Create `packages/simulator`; delete the duplicate; Desktop and Web both consume it. - Reconcile the two `BridgeClient.js` variants into one with mode detection.

Phase 2: Web study mode (3 to 4 weeks) - Content build step producing `exams.json`; schema and linter upgrades (RTP-C-01 to 04). - Study, practice and timed modes in the PWA, with IndexedDB progress (RTP-S-01, 02, 04). - Offline caching of the exam bundle; Lighthouse and axe in CI.

Phase 3: Learning engine (3 weeks) - SM-2 scheduler, weak-area view, flagging, export/import (RTP-S-03). - Question types: multi-select, ordering, lab scenarios linked to the simulator.

Phase 4: Release (2 weeks) - Hosted-runner decision, release workflow, signed artifacts, v0.5 tag. - Desktop hosts the web build; retire Legacy from the gates.

Phase 5: Multi-vendor (later) [A] - Vendor manifest format, second vendor pilot, content contribution guide.

Part 5. Gates (definition of done per phase)

- CI green at the stated head on every leg, with the head SHA quoted in the report.
- Content lint, schema validation and errata tests pass.
- Coverage at or above 70% on Core and the new engine.
- Desktop smoke and PWA offline checks pass.
- Docs counts are generated, not typed.

Part 6. Risks

Risk	Mitigation
Vendor content or trademark complaint	Remove vendor PDFs, add source citations, add disclaimer, confirm posture with Roger
Answer-key errors	Per-question sources, exam-version tags, errata pipeline, reviewer field
CefGlue age	Move Desktop to a maintained WebView host
Scope sprawl (44-sprint plan)	Replace with the five phases above; keep the old plan as an archive
Self-hosted runner dependency	Use hosted runners (free on public repos) unless a reason emerges

Part 7. First actions for Roger’s Claude Code

1. Phase 0 items as separate commits on a work branch.
2. Keep each phase on its own branch with gate evidence in the commit messages. Whether to open PRs is Roger’s call.